Entropic Atlas: Entropy-Guided Spatial Reasoning for Unified Research Agent Benchmarks

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Abstract

Entropic Atlas is a unified research agent that handles two challenging benchmarks through a single Agent-to-Agent (A2A) protocol server: FieldWorkArena, a multimodal spatial question-answering benchmark spanning factory, warehouse, and retail environments, and MLE-Bench, a suite of 75 Kaggle machine learning competitions requiring end-to-end ML engineering. Our key innovation is a structured spatial scene graph engine that extracts entities and relations from vision descriptions, computes distances and safety violations deterministically, then feeds computed facts to large language models — thereby avoiding hallucinated spatial reasoning. Combined with entropy-guided action selection that maximizes information gain per reasoning step and a self-healing ML pipeline with strategy-aware code generation, Entropic Atlas demonstrates a principled approach to building research agents that balance accuracy, cost-efficiency, and robustness across diverse evaluation domains.

1. Introduction

The development of general-purpose research agents capable of operating across diverse evaluation domains represents a fundamental challenge in artificial intelligence. While large language models (LLMs) have demonstrated remarkable reasoning capabilities, deploying them as autonomous agents that can reliably solve real-world tasks remains an open problem. Two recent benchmarks highlight complementary dimensions of this challenge: FieldWorkArena (2025), which evaluates multimodal spatial reasoning in industrial environments, and MLE-Bench (Chan et al., 2024), which tests end-to-end machine learning engineering across 75 Kaggle competitions.

We present **Entropic Atlas**, a unified research agent that addresses both benchmarks through a single A2A protocol server. Our architecture is built on three key contributions:

1. **Spatial Scene Graph Engine**: A structured representation that extracts entities and relations from vision model descriptions, computes spatial relationships deterministically, and produces factual summaries for LLM consumption — eliminating hallucinated spatial reasoning.

2. **Entropy-Guided Reasoning**: An information-theoretic framework that estimates information gain for candidate actions, enabling cost-efficient reasoning by routing queries to appropriate model tiers and triggering reflection only when confidence is low.

3. **Self-Healing ML Pipeline**: A strategy-aware code generation system with automatic error detection, diagnosis, and repair, ensuring robust competition submissions even when initial approaches fail.

2. Related Work

Agent Frameworks. AutoGPT pioneered autonomous LLM agents, while OpenDevin established a software development agent framework with sandboxed execution. SWE-Bench agents demonstrated real-world issue resolution, and DAMO MLE-Agent targets Kaggle-style competitions. Our work unifies two distinct benchmark domains under a single architecture.

Spatial Reasoning in VLMs. Vision-language models exhibit well-documented weaknesses in spatial reasoning, particularly object counting and distance estimation (Liu et al., 2024; Chen et al., 2024). SpatialVLM attempts specialized training, while our approach extracts structured representations and computes spatial facts deterministically.

Scene Graphs. Visual Genome (Krishna et al., 2017) and GQA (Hudson & Manning, 2019) popularized scene graph representations. Our spatial scene graph adapts these ideas to industrial environments with distance computation and constraint checking as first-class operations.

AutoML. AutoGluon (Erickson et al., 2020) and Auto-sklearn automate ML pipelines. Our self-healing pipeline adds strategy-aware code generation and automatic error recovery.

A2A Protocol. Google's Agent-to-Agent protocol defines standard inter-agent communication. Our system implements a compliant A2A server exposing both capabilities through a unified interface.

3. System Architecture

Entropic Atlas operates as a dual-domain A2A server that routes tasks through a domain classifier to the appropriate processing pipeline.

Domain Classification. FieldWorkArena tasks are identified by structured goal format (# Question / # Output Format). MLE-Bench tasks arrive with tar.gz attachments. Classification is deterministic — no LLM call required.

Three-Tier Model Routing:

Tier	Model	Cost (per 1M tokens)	Use Case
Fast	GPT-4.1-mini	$0.40 / $1.60	Classification, parsing
Standard	GPT-4.1	$2.00 / $8.00	Code gen, analysis
Strong	GPT-4.1	$2.00 / $8.00	Spatial reasoning, reflection

Cost Tracking. Each task has a 150K token budget. The cost tracker monitors consumption across all LLM calls, enabling cost-aware routing decisions.

4. Spatial Scene Graph Engine

The spatial scene graph engine is the cornerstone of our FieldWorkArena approach.

Problem. VLMs hallucinate spatial relationships and can't count precisely. Directly prompting with (Image, Question) is unreliable.

Solution — Three-Stage Pipeline:

Stage 1: Entity Extraction. Two-pass: GPT-4.1 vision generates scene description; Florence-2 provides precise bounding boxes and counts.

Stage 2: Graph Construction. Entities become vertices, relations become edges:

- SpatialEntity(id, label, position, attributes, zone)
- SpatialRelation(subject, predicate, object, distance)

Stage 3: Deterministic Computation.

- query_near(v, r): Entities within radius r of entity v
- check_constraints(C): Evaluate safety constraints, return violations
- count_by_label(l): Precise count cross-referenced with Florence-2
- to_fact_sheet(): Serialize graph to structured text for LLM

FieldWorkArena Scoring Functions:

Metric	Description
fuzzy_match	Token-level overlap (threshold 0.8)
exact_match	Case-insensitive string equality
must_include	Answer must contain specified substrings
must_exclude	Answer must not contain specified substrings
json_match	Structured JSON comparison
numerical_match	Numeric comparison (epsilon = 0.05)

5. Entropy-Guided Reasoning

The entropy-guided engine maximizes information gain per reasoning step.

Information Gain. Given candidate actions $\{c_1, \ldots, c_m\}$, select the action maximizing expected reduction in answer entropy: $c^* = \text{argmax}_j \, E[H(A|K_t) - H(A|K_t \cup \text{obs}(c_j))]$

Confidence Estimation. Each answer receives a confidence score $\sigma(a) \in [0, 1]$ via calibrated self-assessment.

Reflection Trigger. When $\sigma(a) < \tau$ ($\tau = 0.6$), reflection is triggered — the agent re-examines reasoning with additional context. Maximum 2 reflection rounds per task.

Algorithm:

1. $a_0, \sigma_0 \leftarrow \text{FastModel}(T, K_0)$
2. if $\sigma_0 \geq 0.8$: return a_0
3. $K_1 \leftarrow K_0 \cup \text{SceneGraph}(T)$
4. $a_1, \sigma_1 \leftarrow \text{StandardModel}(T, K_1)$
5. for $r = 1$ to 2: if $\sigma_1 < \tau$: reflect and retry with StrongModel
6. return a_1

6. Self-Healing ML Pipeline

The MLE-Bench handler transforms competition descriptions into runnable solutions.

Competition Analysis. Auto-detect task type → select strategy template.

Strategy	Task Type	Key Components
Tabular	Classification/Regression	LightGBM/XGBoost, feature engineering
NLP	Text Classification	TF-IDF + LogisticRegression
Vision	Image Classification	Pre-trained CNN, transfer learning
TimeSeries	Forecasting	Lag features, rolling statistics
AutoGluon	Any (default)	AutoGluon TabularPredictor
General	Unknown	Ensemble of lightweight models

Self-Healing Loop:

1. Generate complete Python script from strategy template + LLM
2. Execute in sandboxed subprocess (300s timeout)
3. On failure: parse stderr → generate targeted fix → re-execute
4. Repeat up to 3 times
5. Fallback: dummy submission (always produces valid CSV)

7. Implementation Details

A2A Compliance. Uses a2a-sdk >= 0.3.20. JSON-RPC task submissions with SSE streaming.

Docker Deployment. linux/amd64 container with all dependencies. ENTRYPOINT: uv run src/server.py.

File Processing: Images (JPEG + Florence-2), PDFs (pypdf), Videos (OpenCV frame extraction at 2s intervals), Archives (tar.gz for MLE-Bench), Text (UTF-8).

Resource Budgets: 150K tokens/task, 2 reflection rounds max, 300s code execution timeout, 3 self-healing iterations.

8. Evaluation

FieldWorkArena Ablation Study

Configuration	Factory	Warehouse	Retail
Full System (SSG + EG + F2)	0.72	0.68	0.74
Without SSG (pure VLM)	0.51	0.44	0.55
Without EG (no reflection)	0.65	0.60	0.67
Without F2 (no object det.)	0.63	0.58	0.66
VLM Baseline (GPT-4V)	0.48	0.41	0.52

The spatial scene graph provides +21–24 points over pure VLM. Florence-2 adds +7–10 points. Entropy-guided reasoning adds +7–8 points.

MLE-Bench Results

Category	Valid Submission	Medal Rate	n
Tabular	0.91	0.42	32
NLP	0.78	0.28	18
Vision	0.65	0.15	12
Time Series	0.85	0.35	8
Overall	**0.82**	**0.32**	**75**

Cost Analysis

Domain	Avg. Tokens	Avg. Cost	Avg. Latency
FieldWorkArena	45,200	$0.18	12s
MLE-Bench	92,400	$0.52	180s

9. Discussion

Limitations. Multi-model pipeline introduces ~12s latency per FieldWorkArena task. Quality depends on initial VLM scene description accuracy. Strategy templates are hand-designed for common competition types.

Key Insight. The spatial scene graph provides the largest improvement, confirming that the bottleneck in VLM spatial reasoning is perception reliability, not reasoning ability. Structured representations should be standard in multimodal agent architectures.

Future Work. Fine-tuned Florence-2 for industrial domains; multi-agent collaboration via A2A; streaming responses; expanded benchmarks (SWE-Bench, WebArena).

Broader Impact. Spatial scene graphs have direct applications to industrial safety monitoring. Deployment requires human oversight due to safety-critical implications.

10. Conclusion

Entropic Atlas demonstrates that a unified research agent architecture can address diverse benchmarks through principled design. Our three contributions — spatial scene graphs (+21–24 points), entropy-guided reasoning (+7–8 points), and self-healing ML pipelines (82% valid submission rate) — share a unifying principle: **separate computation from generation**. By computing what can be computed and reasoning only about what must be reasoned about, we achieve improved accuracy and reduced cost compared to end-to-end generative approaches.

Open-sourced at https://github.com/arunshar/entropic-atlas

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